



Capital University of Science and Technology

Department of Computer Science

CS3411 – Operating Systems Lab

Project

Semester: Spring 2026

Dated: 10th June, 2026

Due Date: 17th June 2026

Instructor: Mr. Zohaib Ahmed

Submission Requirements

Each student must submit:

1. Complete source code with proper comments.
2. Project report (3–5 pages) including problem statement, objectives, OS concepts used, screenshots, and conclusion.
3. Demonstration during lab viva.
4. A README file explaining how to compile and run the project.
5. Evidence of implemented OS concepts such as Processes, Threads, Scheduling, IPC, Synchronization, System Calls, or Deadlock Handling.

Evaluation may consider correctness, implementation of OS concepts, code quality, documentation, demonstration, and viva performance.

Following is the list of projects:

Student	Project Title	Description
Shakeel Shafi	Smart Hospital Management Simulator	Simulate a hospital where patients are represented as processes requesting doctors, rooms, and medical equipment. The system should schedule patients based on priority and use synchronization to avoid resource conflicts.
Muhammad Ahmed	Airport Runway Scheduling System	Manage airplane landing and takeoff requests using CPU scheduling algorithms and compare waiting times.
Syed Ahmed Abbas Shah	Multi-Threaded File Compression Tool	Compress multiple files simultaneously using threads and compare performance.
Sharjeel Ahmed Noor	Railway Reservation System	Prevent double booking using synchronization while handling concurrent reservations.
Syed Ali Raza	Distributed Log Processing System	Analyze logs using multiple processes communicating through pipes.
Muhammad Ali	Smart Traffic Signal Controller	Coordinate traffic signals as synchronized threads.
Abdul Mueed	Banking Transaction Processor	Handle concurrent deposits, withdrawals, and transfers safely.
Manahil Saeed	Process Monitoring Dashboard	Display process, CPU, and memory statistics in real time.
Mirza Umer Mustaneer	University Course Registration System	Manage limited course seats during concurrent registrations.
Muhammad Saad Hameed	Restaurant Order Management System	Implement producer-consumer synchronization for restaurant orders.
Muhammad Yousaf Kamal	CPU Scheduling Visualizer	Visualize FCFS, SJF, Priority, and Round Robin scheduling.
Raja Abdul Moiz	Smart Parking Management System	Allocate limited parking slots to vehicle processes.
Muhammad Ahmed	Online Examination Server Simulator	Manage concurrent student access to an examination server.

Ismail Ahmed	Mini Linux Task Manager	Create, suspend, resume, and terminate processes using system calls.
Muhammad Hamza	Cargo Port Container Scheduler	Schedule loading and unloading tasks competing for resources.
Muhammad Haris Altaf	Multi-Threaded Web Server	Serve multiple client requests concurrently using threads.
Ayesha Jamil	Emergency Room Resource Allocator	Allocate hospital resources while preventing deadlocks.
Muhammad Ammar Waheed	Operating System Resource Manager	Manage CPU, memory, printers, and storage resources.
Abdullah Mehmood	Smart Manufacturing Line Simulator	Use IPC and synchronization across production stages.
Muhammad Muzammil Malik	Job Processing Center	Execute jobs using different scheduling strategies.
Ather Ali	Inter-Process Chat Application	Implement real-time communication using IPC mechanisms.
Muhammad Saad	E-Commerce Order Fulfillment System	Coordinate order processing through multiple communicating processes.
Muhammad Fasih-Ur-Rehman	Deadlock Detection and Recovery Simulator	Detect and recover from deadlock situations.
Shahmeer Ahmed Khan Chaudhary	Smart Elevator Control System	Control multiple elevators using threads and scheduling.
Taimoor Haider	Banker's Algorithm Resource Manager	Implement and visualize safe resource allocation.
Syed Awaiz Hassan Kazmi	Cloud Task Scheduling Simulator	Schedule cloud tasks to optimize throughput.
Muhammad Umair	Smart Printer Spooler System	Queue and schedule print jobs from multiple users.
Muhammad Tauqeer-Ul-Hassan	Ride-Hailing Dispatch System	Match drivers and customers using scheduling concepts.
Muhammad Kashif	Multiplayer Game Server Simulator	Synchronize player access to shared game resources.
Abdul Muqeet Rehan	Warehouse Inventory Synchronization System	Maintain consistent inventory with concurrent updates.
Buraq Ahmad Wazir	Smart Data Center Resource Allocator	Optimize server resource allocation and avoid deadlocks.
Mubashir Nawaz	Mini Operating System Simulator	Implement process creation, scheduling, memory management, IPC, and synchronization.